# SMART IRRIGATION AND WATER EFFICIENCY SYSTEM

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## Abstract

A smart irrigation system designed to improve water efficiency in canal-fed agricultural fields. It integrates soil moisture and water level sensors, ESP32 microcontroller, GSM alerts, and solar power to automate irrigation. The system ensures proper water distribution, reduces wastage, and sends real-time notifications to farmers, minimizing manual effort while enhancing productivity in rural farming.

## Problem Statement

- Manual irrigation causes water wastage and uneven field coverage.
- Farmers often miss timely information about water arrival or soil conditions.
- Inefficient irrigation reduces crop productivity and increases labor demand.

## Project Objectives

- Monitor soil moisture and canal water arrival accurately.
- Automate inlet and outlet shutters to control water flow.
- Send real-time GSM alerts for irrigation events and system faults.
- Reduce water wastage and manual supervision.
- Ensure reliable operation with solar-powered energy.

## Methodology

**System Design:** - Square field layout with four corner soil moisture sensors. - Water level sensors at the canal inlet. - Inlet shutter at the front and outlet shutter at the back. - ESP32 microcontroller controls sensors and shutters. - GSM module sends alerts for irrigation start, completion, or failures. - Solar panel provides uninterrupted power. - Optional bird alarm protects crops.

**Process Flow:** 1. Sensors detect water arrival and soil conditions. 2. ESP32 processes data and triggers shutters automatically. 3. GSM module alerts farmer in real-time. 4. Water flows evenly from inlet to outlet, reducing wastage.

**Visuals (to insert 3D model):** - Figure 1: Front view of the field layout with inlet and outlet shutters. - Figure 2: Top view showing sensor placement at field corners. - Figure 3: Perspective view of the automated irrigation system.

## Results & Achievements

- Water and soil conditions monitored effectively.
- Automated shutters ensured precise water control.
- Reduced manual effort and labor cost.
- Minimized water wastage.
- Reliable solar-powered operation.
- Identified areas for scalability and future improvements.

**Visuals (to insert):** - Figure 4: 3D model showing system layout and water flow.

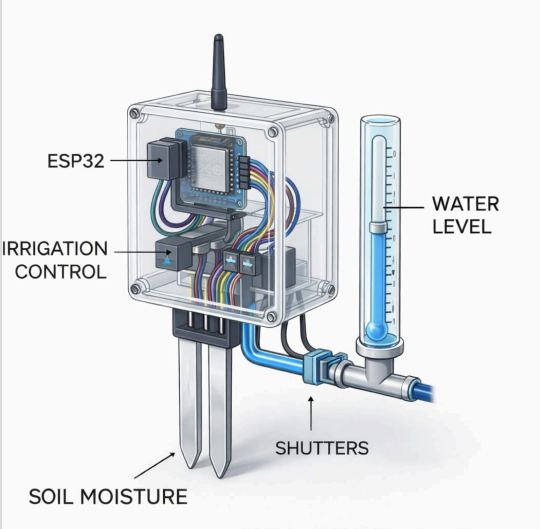
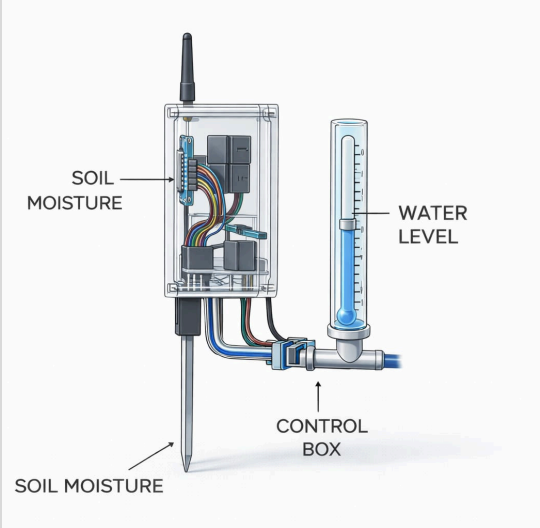
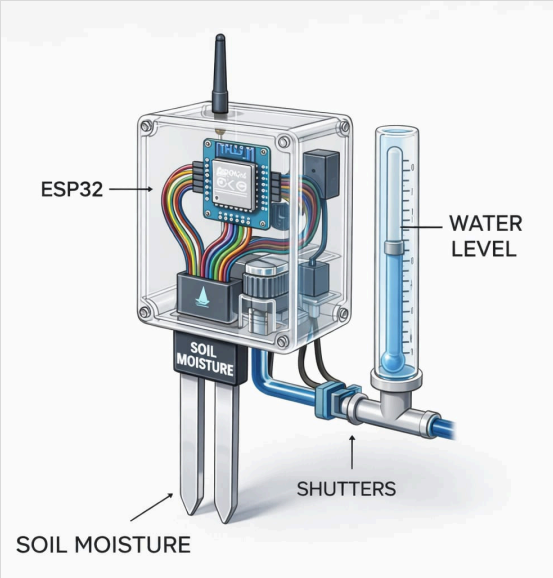
## Business Model & Cost

- **Value Proposition:** Low-cost, reliable, and efficient irrigation system for rural farmers.
- **Target Customers:** Smallholder farmers, agricultural cooperatives, rural development agencies.
- **Revenue Model:** Product sales, installation/service fees, subscription-based alerts.
- **Cost Components:**
  - Sensors (soil moisture & water level)
  - ESP32 microcontroller & GSM module
  - Solar panel and power system
  - Manufacturing, assembly, and deployment
  - Maintenance and support
- **Pricing Strategy:** Affordable for farmers while maintaining sustainable profit.
- **Scalability:** Unit cost decreases with higher adoption and larger production.

## Conclusion

- Developed a feasible, automated smart irrigation system.
- Achieved real-time alert-based monitoring to optimize irrigation.
- Improved water efficiency and reduced manual supervision.
- Strong potential for scalability and real-world deployment in rural areas.

## References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